**Improving Marketing Efficiency Through Customer Segmentation for a UK Online Retailer**

**A Final report for the BDM capstone Project**


Submitted by

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# Declaration Statement

I am working on a Project Title **"Improving Marketing Efficiency Through Customer Segmentation for a UK Online Retailer"**. I extend my appreciation to the **UCI Machine Learning Repository** for providing the publicly available secondary dataset that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from **secondary sources** and has been carefully checked, cleaned, and analysed to ensure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilised for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

**Signature of Candidate:** Adrija Sarkar

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**Date:** 03/01/2026

## 1. Executive Summary

In the rapidly evolving digital marketplace, effective customer segmentation has become a critical driver of marketing efficiency and business growth. This project, titled **“Improving Marketing Efficiency Through Customer Segmentation for a UK Online Retailer,”** applies data-driven techniques to identify meaningful customer segments and develop targeted marketing strategies that optimise revenue potential.

The study is based on transactional data obtained from the **UCI Machine Learning Repository**, representing a **UK-based online retail company** operating between **December 2010 and December 2011**. The clientele consists primarily of wholesalers as well as individual consumers; hence, the business can be described as being active in both B2B and B2C sectors.

After comprehensive data integration and cleaning, the final dataset consisted of **392,692 valid transactions from 4,338 unique customers**, generating a total business revenue of approximately **£8.89 million**.

To understand fundamental business patterns, descriptive and preliminary validation analyses were conducted. These revealed that the **United Kingdom accounts for nearly 82% of total revenue**, confirming the firm’s strong domestic market dependence. Monthly revenue trends indicated **seasonal fluctuations in sales**, while customer-level analysis demonstrated **high inequality in both purchase frequency and spending behaviour**. A moderate positive correlation (**$r \approx 0.55$**) between purchase frequency and monetary value confirmed that repeat buyers typically contribute higher revenue, justifying the application of the **RFM (Recency–Frequency–Monetary) framework**.

Using RFM analysis, customers were segmented based on:

- **Recency:** Time since last purchase
- **Frequency:** Number of transactions
- **Monetary:** Total revenue contribution

After standardising the RFM variables, the **Elbow Method** was applied to determine the optimal number of customer segments, resulting in **$k = 3$**, which was implemented using the **K-Means clustering algorithm**. This produced three highly interpretable customer segments:

1. **At-Risk / Low-Engagement Customers (1,082 customers):**
   Characterised by high recency, low frequency, and low spending, these customers exhibit weak engagement and high churn probability.

2. **Regular / Core Revenue Customers (3,230 customers):**
   This largest customer group displays moderate recency, consistent purchasing behaviour, and stable revenue generation, forming the operational backbone of the business.

3. **VIP / Champion Customers (26 customers):**
   Although small in number, this elite group demonstrates extremely recent activity, exceptionally high purchase frequency, and extraordinary spending levels, making them strategically critical to revenue sustainability.

Based on these insights, **segment-specific marketing strategies** were proposed. At-risk customers require low-cost reactivation campaigns, regular customers benefit from loyalty programs and personalised promotions, while VIP customers demand high-touch retention strategies such as exclusive offers, early access, and premium relationship management. This segmentation-driven approach enables the firm to transition from uniform marketing to **value-based targeting**, thereby improving **marketing efficiency, customer retention, and long-term profitability**.

Overall, this project demonstrates the successful application of business data management and machine learning techniques to solve a real-world retail problem. The findings clearly establish that **data-driven customer segmentation is a powerful strategic tool** for enhancing marketing effectiveness and sustaining competitive advantage in the online retail sector.

## 2. Proof of Originality

- The data has been collected from UCI Machine Learning repository: https://archive.ics.uci.edu/dataset/352/online+retail

- The additional details were collected from an associated academic article: "Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining "

The complete data cleaning, exploratory analysis, feature engineering, RFM computation, clustering, and visualisation workflow for this project was implemented in Python using Google Colab. The analysis code is fully reproducible and has been made publicly accessible for validation purposes.

- Google Colab Notebook (view-only): https://colab.research.google.com/drive/1Nl8qCYID-iqFPJLykGg26MStfBY9uSr0?usp=sharing

- GitHub Repository (code and supporting files): https://github.com/codeadri/bdm-customer-segmentation

The repository contains the (final analysis notebook, supporting data files) documentation required to reproduce all results and figures presented in this report.

## 3. Meta data and descriptive statistics

The dataset used in this study is the Online Retail Dataset, obtained from the UCI Machine Learning Repository. It contains real transactional data of a UK-based non-store online retail company that primarily sells unique all-occasion gift items. The business operates in both business-to-consumer (B2C) and wholesale (B2B) segments.

The dataset covers a one-year transactional period from December 2010 to December 2011, capturing detailed records of customer purchases, product information, pricing, order quantities, and the geographic location of customers (541,909 instances).

After data integration and cleaning, the final analytical dataset consists of:

- 392,692 valid transaction records
- 4,338 unique customers
- Total business revenue of £8.89 million

The dataset represents multivariate, time-series transactional data, making it suitable for behaviour-based customer segmentation and purchase pattern analysis.

| Variable Name | Description |
|---|---|
| **InvoiceNo** | Unique identifier for each transaction |
| **StockCode** | Unique product/item code |
| **Description** | Description of the product/item |
| **Quantity** | Number of units purchased per transaction |
| **InvoiceDate** | Date of transaction |
| **InvoiceTime** | Time of transaction |
| **InvoiceDateTime** | Combined transaction timestamp used for time-based analysis |
| **CustomerID** | Unique identifier for each customer |
| **Country** | Country of customer residence |
| **UnitPrice** | Price per unit of the product |
| **Revenue** | Total transaction value (Quantity × UnitPrice) |

**CustomerID** is treated strictly as a categorical identifier and is used only for aggregation purposes; it is not used as a numerical variable in any statistical or clustering computations.

**Customer-Level Descriptive Statistics:**

To understand customer purchasing behaviour, transactional data was aggregated at the customer level. Descriptive statistics were then computed for purchase frequency and total monetary value to assess central tendency and dispersion.

| | **Purchase Frequency** | **Total Spend** |
|---|---|---|
| **Mean** | 4.27 | 2048.69 |
| **Median** | 2.00 | 668.57 |
| **Standard deviation** | 7.70 | 8985.23 |
| **Minimum** | 1.00 | 3.75 |
| **Maximum** | 209.00 | 280206.02 |

The findings show a great deal of variation in consumer behaviour, with standard deviations for both frequency and monetary value considerably higher than the mean. This demonstrates a highly skewed distribution where a tiny portion of the clientele makes a disproportionate contribution to overall revenue. The use of behaviour-based segmentation methods like RFM analysis is justified by these features.

## 4. Detailed Methodology

### 4.1. Data Collection:

The dataset used for this study was obtained from the UCI Machine Learning Repository and consists of real transactional records of a UK-based online retail company operating in both B2C and B2B segments. The dataset spans transactions recorded between December 2010 and December 2011 and includes information on invoices, products, quantities, customer identifiers, and geographic location. Initially, the dataset was downloaded in Excel format. It contained key transactional identifiers such as InvoiceNo and StockCode, but was found to be missing the UnitPrice attribute. To ensure data completeness, the same dataset was also imported into a Google Colab environment using the **ucimlrepo** Python library. This version included UnitPrice but did not explicitly expose InvoiceNo and StockCode. Therefore, a hybrid integration approach was adopted to reconstruct a complete transactional dataset.

### 4.2. Data Integration and Dataset Assimilation

Before merging the datasets, the original *InvoiceDate* column from the Excel dataset was split into two separate variables:

- **InvoiceDate** (dd-mm-yyyy format)
- **InvoiceTime** (hh:mm:ss format)

This step enabled accurate transaction-level timestamp creation. The modified Excel dataset containing *InvoiceNo, StockCode, Description, Quantity, InvoiceDate, InvoiceTime, CustomerID,* and *Country* was imported into Python as a pandas DataFrame. The *UnitPrice* variable was then extracted from the Python-imported UCI dataset and merged into the Excel-based DataFrame using common transactional attributes. This process resulted in a fully reconstructed dataset containing all essential business variables required for subsequent analysis.

### 4.3. Creation of Transaction Timestamp

For precise recency measurement and time-based analysis, *InvoiceDate* and *InvoiceTime* were combined into a single timestamp variable named InvoiceDateTime using pandas datetime conversion. This consolidated timestamp served as the temporal foundation for RFM analysis and time-trend evaluation.

### 4.4. Data Cleaning Procedures Applied:

To ensure analytical reliability, a structured data cleaning process was applied:

- **Cancelled Transaction Removal:** All invoice numbers beginning with the letter “C”, representing cancelled orders, were removed.
- **Invalid Quantity Filtering:** Transactions with zero or negative quantities were excluded to eliminate returns and erroneous sales entries.
- **Invalid Price Filtering:**
  Transactions with zero or negative unit prices were removed to ensure the correctness of revenue calculations.

- **Missing Customer Identifier Removal:**
  Records with missing *CustomerID* values were dropped, as customer-level segmentation is not possible without unique identifiers.
- **Duplicate Removal:**
  Exact duplicate records were eliminated to avoid overestimation of transactional activity.

### 4.5. Feature Engineering (Revenue Creation):

A new business-critical variable, Revenue, was created as:

$$\textbf{Revenue} = \textbf{Quantity} \times \textbf{UnitPrice}$$

This variable represents the total value generated per transaction and forms the monetary basis for the RFM model.

### 4.6. Final Clean Dataset Output:

After all cleaning and integration steps, the final dataset comprised:

- 392,692 valid transactional records
- 4,338 unique customers
- Total business revenue of £8,887,208.894

The final variables included are *InvoiceNo, StockCode, Description, Quantity, InvoiceDate, InvoiceTime, CustomerID, Country, UnitPrice, InvoiceDateTime,* and *Revenue*.

### 4.7. RFM-Based Customer Segmentation Framework:

To determine how customers behave and categorise them accordingly, the Recency–Frequency–Monetary (RFM) framework was employed. The RFM model is a popular method in retail and marketing, and customers are sorted based on their buying habits using three key factors: how recently they bought, how often they buy, and how much money they spend.

**Recency** is used to tell how long it's been since a customer last bought something. It's a quick way to see how engaged they are now and if they might be about to stop buying (churn risk). **Frequency** is the total number of purchases a customer has made over a set time, showing how regular they are and how loyal they might be. **Monetary value** is the total amount of money that has been spent by a customer, which shows how much they are worth to the business overall.

The RFM framework was chosen because it directly uses transaction data to show customer engagement, loyalty, and how much revenue is brought in, making it perfect for retail segmentation when personal information like demographics is not available. Unlike grouping customers by who they are or what they think, RFM focuses only on what they *do*, allowing customers to be grouped based purely on objective, data-driven economic value.

After the transaction data was cleaned up, the RFM scores were calculated for each customer. Recency was simply the number of days between the customer's last purchase and the very latest date in the entire dataset. Frequency was defined as the count of unique transactions (invoices) linked to each customer, and monetary value was the total money that had been spent by the customer. These scores became the main data points (features) for the next step, the clustering analysis.

### 4.8. K-Means Clustering for Customer Group Identification:

An unsupervised machine learning approach, specifically K-Means clustering, was applied to the constructed RFM metrics to identify distinct customer segments. K-Means was chosen for its computational efficiency, interpretability, and effectiveness with continuous numerical features like the RFM variables.

Prior to clustering, the RFM variables underwent standardisation using z-score normalisation. This essential step ensured that all variables contributed equally to the distance-based K-Means algorithm, preventing high-magnitude monetary values from dominating the cluster formation. Since K-Means relies on Euclidean distance, unscaled features could have otherwise biased the results toward variables with larger numerical ranges.

The optimal number of clusters was determined using the Elbow Method, which analysed the within-cluster sum of squared errors (SSE) for various $k$ values. The resulting 'elbow point' at $k = 3$ indicated the best balance of model simplicity and explanatory power. Consequently, the final K-Means model was trained to produce three clusters.

This clustering process successfully yielded three distinct and interpretable customer segments, each defined by a unique RFM profile. These segments were then analysed to understand differences in customer behaviour and to formulate specific marketing strategies. The derived clusters serve as the fundamental analytical basis for the findings and recommendations presented later in this report.

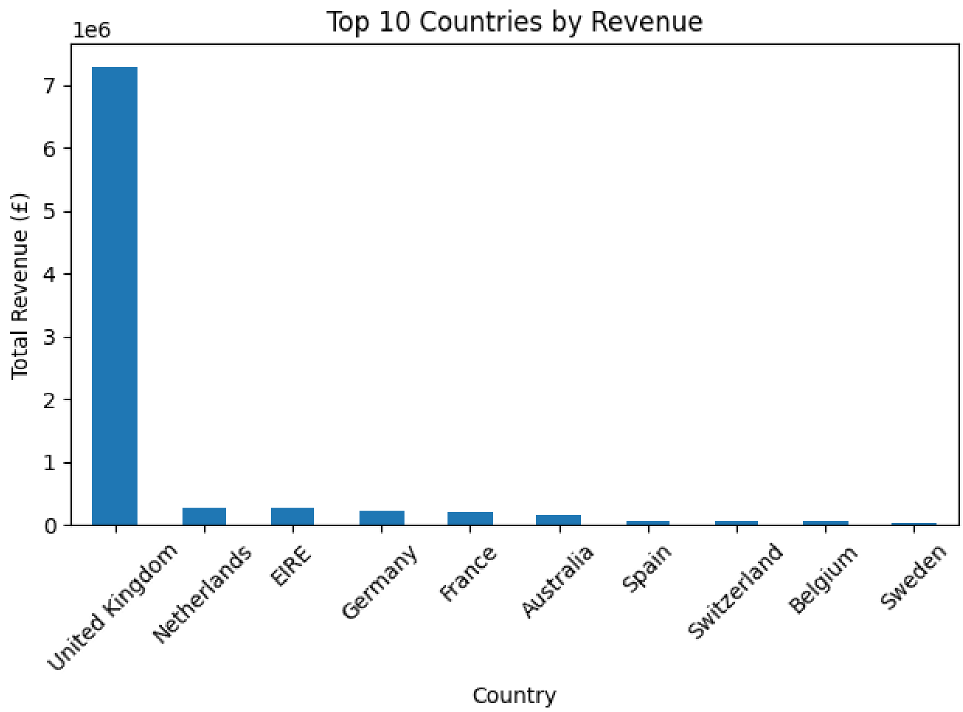
# 5. Results and findings

## 5.1. Overall Business Performance:

- Total Transactions: 392,692
- Unique Customers: 4,338
- Total Revenue: £8,887,208.894
- Average Order Value: £479.56

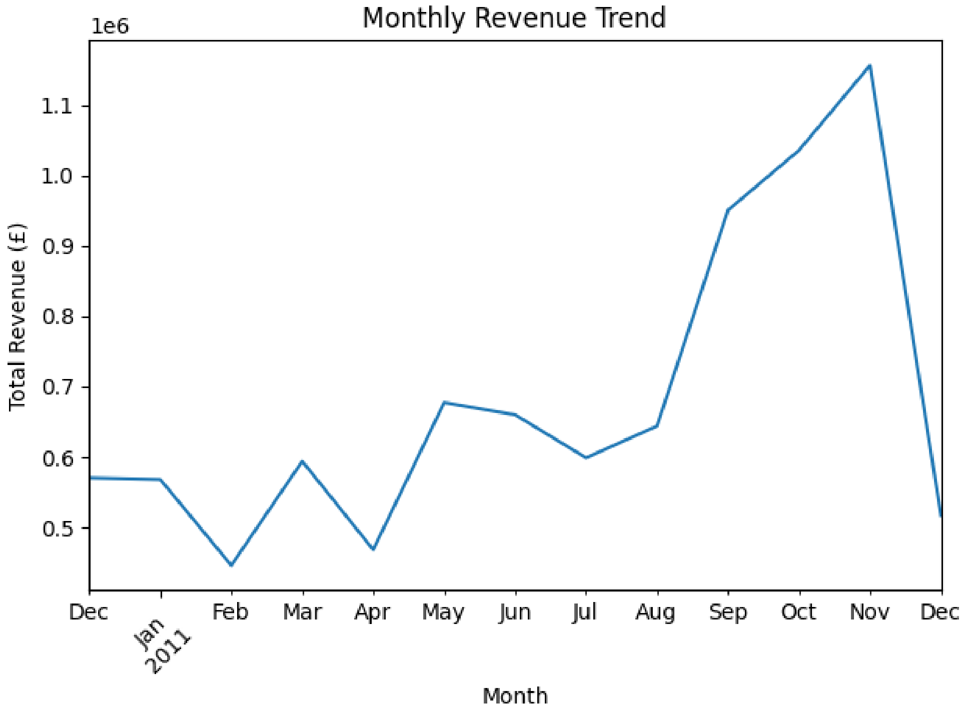
These indicators confirm the presence of a high-volume, high-value online retail operation.

## 5.2. Country-Wise Revenue Distribution




The country-wise revenue graph shows that the United Kingdom contributes approximately 82% of total revenue, confirming that the business is primarily UK-centric. Secondary markets include the Netherlands, EIRE, Germany, and France, each contributing between 2–3% of total revenue.

### 5.3. Monthly Revenue Trend




Monthly revenue analysis reveals noticeable temporal variations across the year, indicating the presence of seasonality and demand cycles. Revenue peaks during festive and promotional periods, supporting the importance of time-based marketing strategies.

### 5.4. Customer Purchase Frequency Distribution

| Customer ID | Purchase Count |
|---|---|
| 12748 | 209 |
| 14911 | 201 |
| 17841 | 124 |
| 13089 | 97 |
| 14606 | 93 |
| 15311 | 91 |
| 12971 | 86 |
| 14646 | 73 |
| 16029 | 63 |
| 13408 | 62 |

The frequency distribution (consisting top 10 CutomerID) shows that:

- The median customer makes only 2 purchases
- The maximum observed frequency is 209 purchases

This indicates a highly skewed purchasing pattern, where a small group of customers contributes disproportionately to repeat business.

### 5.5. Customer Monetary Value Distribution (top 10)

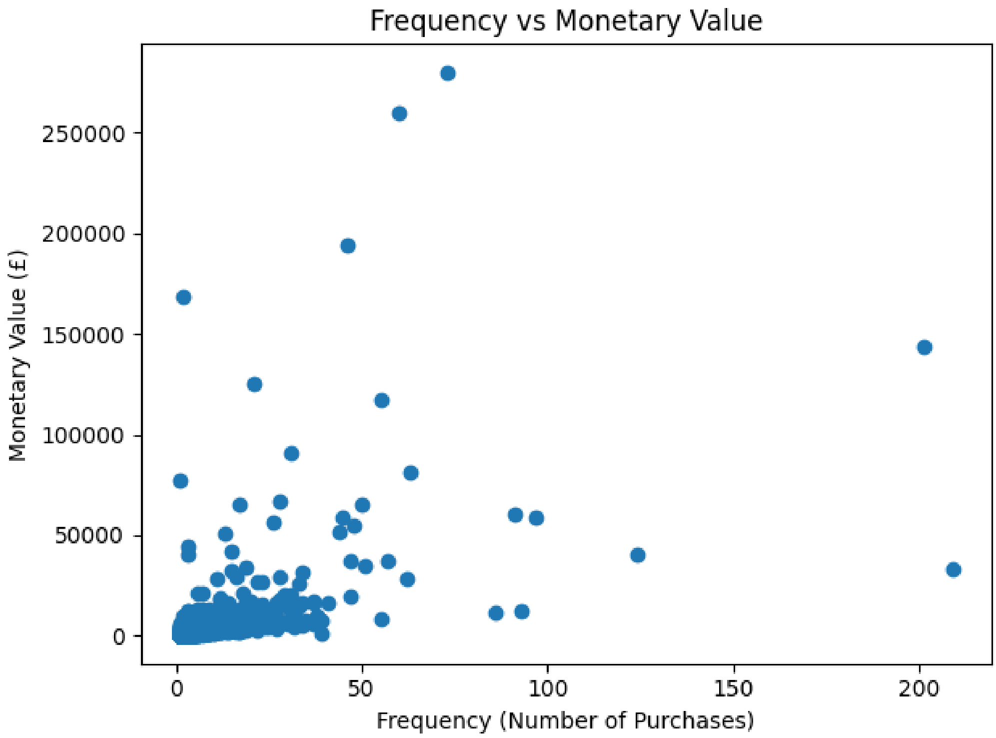
| Customer ID | Revenue |
|---|---|
| 14646 | 280206.02 |
| 18102 | 259657.30 |
| 17450 | 194390.79 |
| 16446 | 168472.50 |
| 14911 | 143711.17 |
| 12415 | 124914.53 |
| 14156 | 117210.08 |
| 17511 | 91062.38 |
| 16029 | 80850.84 |
| 12346 | 77183.60 |

Customer monetary value shows:

- Median spend ≈ £669
- Maximum spend ≈ £280,206

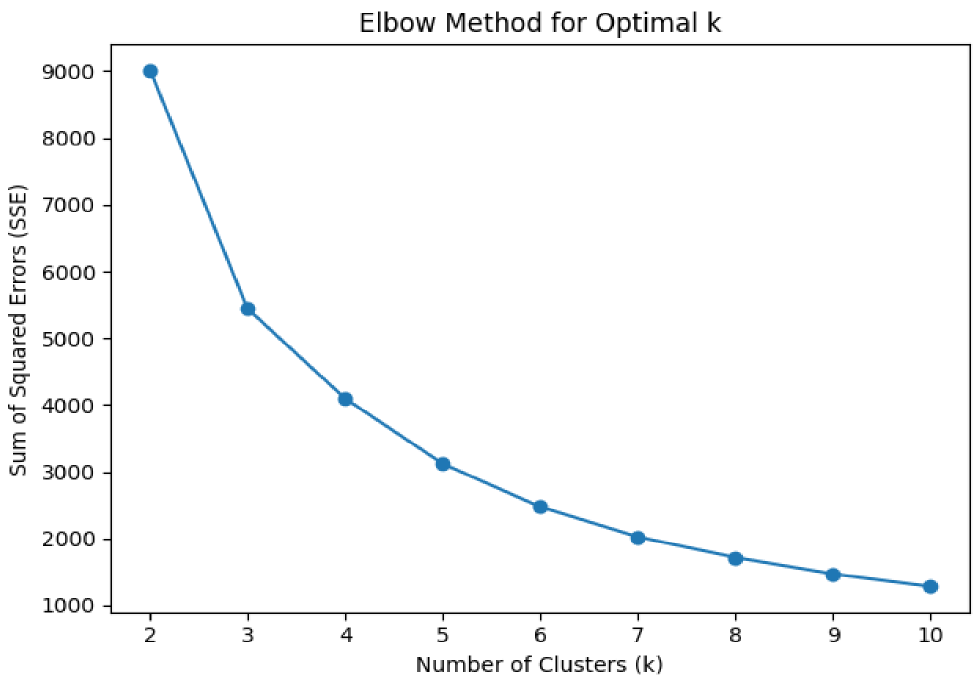
This confirms extreme revenue concentration, where a small number of customers contribute a very large share of overall revenue.

### 5.6. Frequency vs Monetary Relationship




The scatter plot demonstrates a **moderate positive correlation (≈ 0.55)** between Frequency and Monetary value, confirming that frequent buyers generally generate higher revenue. This supports the behavioural logic underlying RFM segmentation.

### 5.7. Elbow Method for Optimal k



The Elbow curve shows a sharp decrease in SSE up to **k = 3**, after which the curve flattens. This confirms that **three clusters provide an optimal balance between segmentation quality and interpretability**.

### 5.8. Final Customer Cluster Profiles

Cluster-level averages for Recency, Frequency, and Monetary value are summarised below:

| Cluster | Recency | Frequency | Monetary (£) | Customers |
|---|---|---|---|---|
| 0 | 247.11 | 1.58 | 629.66 | 1,082 |
| 1 | 41.45 | 4.67 | 1,849.67 | 3,230 |
| 2 | 6.04 | 66.42 | 85,826.08 | 26 |

These clusters exhibit clearly differentiated behavioural profiles consistent with the RFM framework.

- **Cluster 0 (Low-Engagement / At-Risk Customers):**
  Customers in Cluster 0 have the **highest recency (247 days)**, indicating that a long period has passed since their last transaction. Their **low purchase frequency (1.58)** and **low average monetary value (£629.66)** suggest weak engagement with the business and limited lifetime value. From an RFM perspective, this cluster scores poorly across all three dimensions, reflecting customers who are inactive and highly susceptible to discontinuation. Their purchasing behaviour suggests they made one-time or infrequent purchases and failed to develop repeat-buying habits.

- **Cluster 1 (Regular / Core Revenue Customers):**
  Cluster 1 contains the majority of customers (3,230) and represents the firm's operational customer base. These customers display **moderate recency (41 days)**, indicating relatively recent interaction with the retailer. Their **frequency (4.67)** and **monetary contribution (£1,849.67)** are substantially higher than Cluster 0 but far lower than Cluster 2. In RFM terms, these customers exhibit balanced engagement and stable purchasing behaviour. They consistently contribute to revenue but do not exhibit excessive loyalty or spending concentration. This segment is the foundation of routine business performance and provides the best opportunity for incremental revenue growth via retention and upselling strategies.

- **Cluster 2 (VIP / Champion Customers):**
  Cluster 2 represents an elite segment of only 26 customers, yet it demonstrates **exceptionally low recency (6 days)**, extremely high **frequency (66 purchases)**, and an extraordinary **monetary value (£85,826)**. These values indicate that these customers make frequent purchases, spend significantly more per customer, and have

had recent interactions with the company. Under the RFM framework, this cluster scores highest across all three dimensions and therefore represents the most valuable customers in terms of lifetime value and revenue dependency. Although numerically small, this cluster contributes a disproportionately large share of total revenue, making it strategically critical to the business.

Overall, the cluster profiles align strongly with the behavioural logic of RFM segmentation. The results confirm the existence of three distinct behavioural groups: disengaged customers with high churn risk, stable repeat customers forming the core business, and a small group of high-value customers that drive a substantial portion of revenue.

## 6. Interpretation of results and recommendation

The clustering results provide clear evidence that customer behaviour within the retailer's customer base is highly heterogeneous. Differences in recency, purchase frequency, and monetary contribution demonstrate that customers interact with the business in fundamentally different ways. Consequently, a single uniform marketing strategy is inefficient, as it fails to account for variations in customer value and engagement.

### 6.1. Cluster 0 – At-Risk / Low-Engagement Customers (1,082 Customers)

Customers in this cluster have long periods of inactivity, minimal purchase repetition, and low total spending. This behavioural profile indicates low brand attachment and a high likelihood of churn. It would be economically inefficient to invest heavily in customer retention without first reactivating them.

- **Recommended Strategies:**
  Implement low-cost reactivation campaigns such as reminder emails and limited-time discounts.
- Use automated remarketing techniques rather than personalised high-cost campaigns.
- Focus on identifying which customers respond to reactivation efforts before allocating additional resources.
- Avoid premium loyalty incentives unless renewed engagement is observed.

The objective for this segment is behavioural reactivation rather than immediate revenue maximisation.

### 6.2. Cluster 1 – Regular / Core Revenue Customers (3,230 Customers)

This cluster exhibits consistent purchasing behaviour, with moderate recency and a steady monetary contribution. These customers are the firm's most consistent revenue source and the most responsive to marketing interventions.

**Recommended Strategies:**

- Introduce structured loyalty programs to reinforce repeat purchasing.
- Apply cross-selling and upselling strategies based on historical purchase patterns.
- Use personalised product recommendations to increase basket size.
- Provide periodic promotional offers to prevent migration into the at-risk cluster

The primary strategic goal for this group is long-term retention and gradual revenue expansion through relationship strengthening.

### 6.3. Cluster 2 – VIP / Champion Customers (26 Customers)

This cluster shows extremely recent purchases, very high transaction frequency, and exceptionally high spending. These customers are responsible for a disproportionate share of total revenue and therefore represent a major financial risk if lost.

**Recommended Strategies:**

- Provide exclusive product previews and early access to sales.
- Offer premium loyalty benefits and customised incentives.
- Assign dedicated relationship management or priority service.
- Focus on retention rather than price-based discounting.

For this segment, the objective is revenue protection rather than acquisition or price competition.

### 6.4. Strategic Impact of Segmentation

The segmentation framework enables the retailer to transition from mass marketing to value-based marketing. By aligning marketing actions with behavioural customer value, the firm can:

- Improve marketing return on investment (ROI) by allocating resources more efficiently.
- Reduce churn among low-engagement customers through timely intervention.
- Strengthen loyalty among core customers.
- Protect high-value revenue streams generated by VIP customers.

Overall, the segmentation-driven strategy transforms customer management from a reactive model to a proactive and data-driven system. This allows marketing decisions to be guided by objective behavioural evidence rather than intuition, thereby supporting sustainable long-term growth and improved marketing efficiency.

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